Zachary Randall, Florida Museum (UF 236274) ©

Misgurnus anguillicaudatus (Cantor, 1842)

Common name: Pond Loach

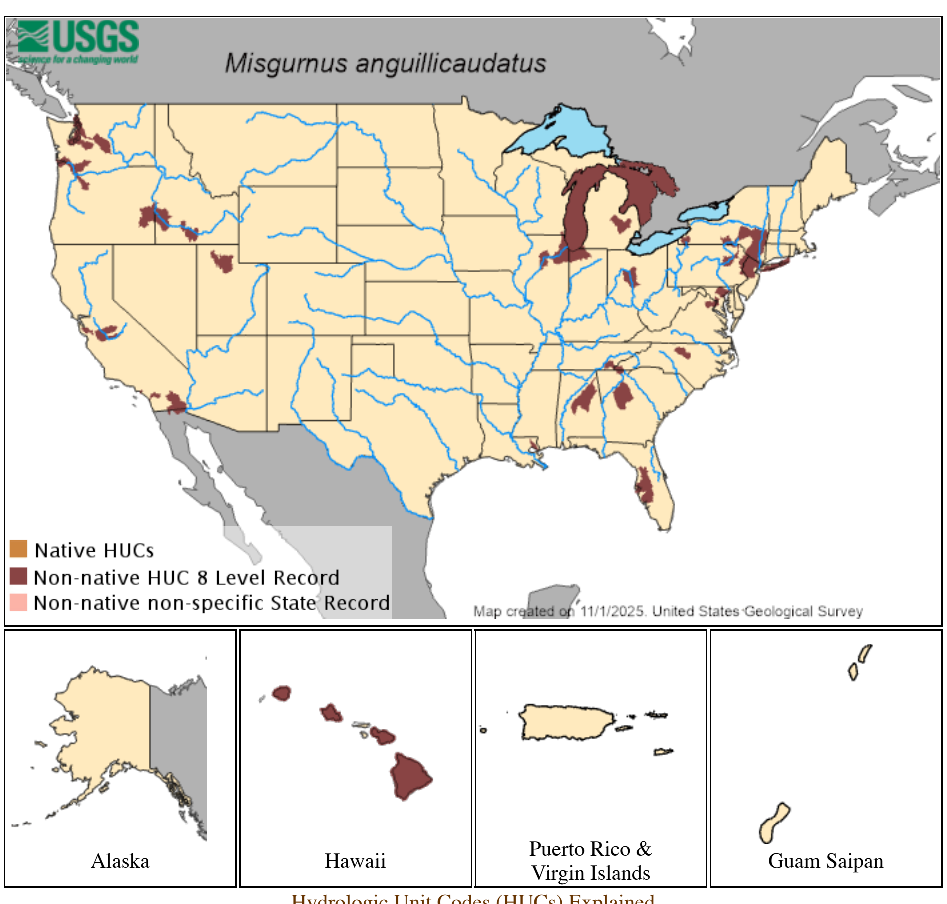
Synonyms and Other Names: Oriental Weatherfish, Dojo, Dojo Loach, Weather Loach, Japanese Weatherfish, Amur Weatherfish

Taxonomy: available through ITIS

Identification: Eel-like body, brown with greenish grey-brown marble markings dorsally, pale silver ventrum. The mouth is small, narrow and subinferior surrounded by six barbels. The lips are thick and fleshy. Short lateral line, not exceeding the length of the pectoral fin. Has a stout spine on its pectoral fin. Dorsal fin origin is above the pelvic fin origin. There are 9 dorsal rays, 6–7 pelvic rays, and 7–8 anal rays. Distinguishing characteristics were given by Berg (1949), Sterba (1973), Masuda et al. (1984), Page and Burr (1991), and Talwar and Jhingran (1992). The species is included in identification keys provided by Nichols (1943), Berg (1949), Moyle (1976), Talwar and Jhingran (1992), and Laird and Page (1996). Illustrations or photographs appeared in Nichols (1943), Berg (1949), Masuda et al. (1984), and Axelrod et al. (1985).

Size: To 28 cm SL (Kottelat and Freyhof 2007), but averages 10-20 cm.

Native Range: Eastern Asia from Siberia south to northern Vietnam, including Japan (Berg 1949; Masuda et al. 1984; Talwar and Jhingran 1992).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Misgurnus anguillicaudatus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	2009	2025	3	Cahaba; Lower Coosa; Middle Coosa
CA	1963	2025	7	Coyote; Los Angeles; Middle San Joaquin-Lower Chowchilla; Salton Sea; Santa Margarita; Seal Beach; Southern California Coastal
FL	1988	2025	7	Alafia; Hillsborough; Little Manatee; Myakka; Peace; Tampa Bay; Withlacoochee
GA	2020	2025	2	Upper Ocmulgee; Upper Oconee
HI	1870	2022	4	Hawaii; Kauai; Maui; Oahu
ID	1985	2023	3	Lower Boise; Middle Snake-Payette; Upper Snake-Rock
IL	1987	2023	5	Chicago; Des Plaines; Kankakee; Lower Illinois-Senachwine Lake; Upper Illinois
IN	2002	2023	3	Chicago; Kankakee; Little Calumet-Galien
LA	2005	2005	1	Liberty Bayou-Tchefuncta
MD	2009	2009	1	Gunpowder-Patapsco
MI	1939	2000	3	Flint; Lake Huron; Shiawassee
NJ	2007	2025	3	Hackensack-Passaic; Middle Delaware-Musconetcong; Raritan
NY	2001	2025	9	Bronx; Conewango; Hudson-Wappinger; Middle Delaware-Mongaup-Brodhead; Middle Hudson; Rondout; Schoharie; Southern Long Island; Upper Susquehanna
NC	2009	2019	1	Haw
OH	2019	2025	1	Upper Scioto
OR	1991	2017	8	Clackamas; Lower Columbia; Lower Columbia-Clatskanie; Lower Malheur; Lower Owyhee; Lower Willamette; Middle Snake-Payette; Middle Snake-Succor
PA	2017	2024	3	Owego-Wappasening; Upper Susquehanna; Upper Susquehanna-Lackawanna
TN	1995	1995	1	Hiwassee
UT	2015	2025	3	Great Salt Lake; Jordan; Lower Weber
VA	2019	2019	1	Middle Potomac-Anacostia-Occoquan
WA	1978	2024	5	Lake Washington; Lewis; Puget Sound; Puyallup; Upper Yakima
WI	2000	2000	1	Lake Michigan

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: The Pond Loach is found in muddy or silty substrates in low-gradient, shallow water, often in aquatic macrophyte beds (Tabor et al. 2001; Schmidt and Schmidt 2014). It does not forage by sight, but requires chemical stimuli to induce feeding behavior (Watanabe and Hidaka 1983). Its diet is composed mostly of small benthic invertebrates and detritus; Tabor et al. (2001) found that cladocerans and chironomids were the most frequently ingested prey items collectively; Schmidt and Schmidt (2014) found that amphipods, gastropods and chironomids comprised the most important food items in several New York populations; Urquhart and Koetsier (2014b) identified chironimids and small bivalves as the most frequently ingested prey items in Idaho populations.

Pond Loach is highly tolerant to extreme conditions or marginal habitat quality: Koetsier and Urquhart (2012) found that Pond Loach can withstand desiccation and starvation for more than 81 days, and Urquhart and Koetsier (2014a) estimated a critical thermal minimum of between -1° and -2°C and reported survival of experimental fish partially or fully encased and in direct contact with ice. Further, it can withstand a maximum temperature of around 38°C and O2 concentrations below 2 mg/L (McMahon and Burggren 1987; Rixon et. al. 2005; Urquhart and Koetsier 2014b). It can use its intestine as an accessory respiratory organ, enabling it to live in oxygen-poor waters and to bury itself in soft substrates to survive long droughts (Ip et al. 2004). As a side effect of its ability to breathe air, Pond Loach can also tolerate extremely high concentrations of ammonia (Ip and Chew 2018).

This species exhibits sexual size dimorphism, with the average length of female Pond Loach being considerably larger than that of the males (Tabor et al. 2001; Schmidt and Schmidt 2014). It is an external fertilizer that spawns multiple times from April to October (Milton et al. 2018). Schmidt and Schmidt (2014) examined the fecundity of introduced New York populations: ovary weight was 4-19% of total wet body weight, and total oocytes in unmated females ranging between 150-18,000 (with fecundity increasing with total length).

Means of Introduction: California populations are apparently descended from individuals that escaped from a local goldfish farm, possibly as early as the 1930s (St. Amant and Hoover 1969). Similarly, populations in the upper Shiawassee River system, Michigan, are believed to have descended from individuals imported from Kobe, Japan, which escaped from a nearby aquarium supply company in Oakland County before 1951, possibly as early as 1939 (Schultz 1960; Courtenay et al. 1984). In Hawaii, the species was presumably introduced by Asian immigrants during the 1800s, probably for food (Maciolek 1984; Devick 1991). Brock (1960) stated that its use as bait for fish was apparently instrumental in its spread in Hawaii. Collections in Louisiana were the result of escapes from a fish farm. Introductions into other states were presumably the result of aquarium releases.

Status: Established in Shiawassee River and Lake Michigan (Simon et al. 2006; USEPA 2008). Established in Alabama, California, Florida, Georgia, Hawaii, Idaho, Illinois, Michigan, New Jersey, New York, Oregon, and Washington, and possibly in Louisiana. Status is unknown in Tennessee, North Carolina, and Maryland.

Great Lakes: Overwintering and reproducing in Lake Michigan.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological | Economic | Human Health | Other

It may reduce populations of aquatic insects important as food to native fishes (Page and Laird 1993; USEPA 2008). Based on its preferred habitat and diet, and on its population size, Maciolek (1984) categorized *Misgurnus* along with several other introduced fishes as species having an intermediate impact on Hawaiian streams. Schmidt and Schmidt (2014) suggested that Pond Loach may compete with native Eastern Mudminnow (*Umbra pygmaea*) for food where the two species co-occur. Experimentation in Australia on the environmental impacts of Pond Loach suggests it may significantly reduce macroinvertebrate abundance and increase turbidity and nitrogen levels in standing water (Keller and Lake 2007).

Remarks: In addition to appearing in the aquarium trade, Pond Loach has been introduced into several parts of the world for aquaculture and as a bait fish (Welcomme 1988). It is considered a cool-water species (Welcomme 1988); individuals burrow into soft substrates and tolerate low-oxygen conditions by breathing atmospheric air (Sterba 1973). As its name suggests, this fish becomes more active in response to changes in barometric pressure; thus it frequently has been cited as a harbinger of storms (Sterba 1973; Axelrod et al. 1985). There is concern that if Pond Loach becomes more abundant and spreads, it will reduce populations of aquatic insects important as food for native fishes (Page and Laird 1993). Maciolek (1984) categorized *Misgurnus*, along with several other introduced fishes, as having an intermediate impact on Hawaiian streams based on their preferred habitat and diet and on their numbers. The Bolsa Chica Channel, California, cited by Shapovalov et al. (1981), likely represents the Huntington Beach, Orange County site reported by other authors (e.g., Courtenay et al. 1986; Page and Burr 1991). Shafland et al. (2008) reclassified this species from Established to Possibly Established in Florida because of unreliability of collection of specimens.

Voucher specimens: **California** (CAS 24192; LACM 36986.002, 36988.002, 56916.001; SIO 68-213; UF 87981); **Florida** (UF 98633, 98634, 100519, 100533, 119925, 128570, 139432, 139507, 143225, 143230, 143271, 143299, 143309, 143376, 143381, 143386, 143541, 143728, 143738, 148188, 148224, 163658, 163662, 163673, 163681, 163684, 163694, 163695, 163719, 163751, 163777); **Hawaii** (BPBM 3624, discarded in 1969; ANSP 54151, 89309; UWFC 115671); **Idaho** (UNLV 1951); **Illinois** (INHS 61129, 61130, 61131); **Michigan** (UMMZ 173823, 174568, 198978, 212412, 225614); **New Jersey** (ANSP 188939); **New York** (NYSM 53924, 55379, 57313, 54195, 65640, 65919, 65922, 68732, 68734); **Oregon** (OS 13196, 13215, 13216, 15467, 15473, 15474; Portland State University); **Tennessee** (UT 46.17); **Washington** (UF 168231; UWFC 111106)

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Other Resources:

The Great NANFA Loach Hunt of 1999!

Pond Loach in Idaho

Great Lakes Waterlife

FishBase Summary

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